

Ádám Fodor

MACHINE LEARNING ENGINEER

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PROFILE

A decade of building and shipping ML systems: multimodal deep learning systems carried from design through evaluation and optimization to **deployment**, then kept working as released, versioned packages. **Nine PyPI packages**, two deployed services with public demos, **ten peer-reviewed publications** including two **Q1** journals.

EXPERIENCE

Research Scientist, Neural Information Processing Group (NIPG) Sep 2017 – Present
Eötvös Loránd University, Faculty of Informatics, Dept. of Artificial Intelligence · Budapest, Hungary

- **Designed LinMuT**, a general-purpose multimodal sequence model with **linear-complexity attention**, then adapted it across three projects, each evaluated **cross-dataset**: blink **event detection**; personality and behavior estimation; emotion estimation with **multi-task learning across disjoint datasets**.
- **Shipped an eye blink detector end to end**: model, **Q2** journal paper, PyPI package, real-time inference. Then ported it to *Tobii* head-mounted camera video and integrated it into an ADOS-2 autism behavior-analysis framework delivered to **Rush University Medical Center** and published in a **Q1** journal.
- **Split the runtime, PyTorch for development and ONNX for release**: the blink detector's released package drops torch entirely, 469 MB smaller and 4.15 ms/frame instead of 20.19; every export is gated on numerical parity probes against the *PyTorch* graph so the two paths do not drift apart.
- **Put the personality model on Google Cloud Run**: *MLflow registry* as the promotion boundary, Optuna hyperparameter search, ONNX model served from Docker, scale-to-zero so an idle demo costs nothing.
- **Back metric gains with significance tests**: paired bootstrap, 95% CIs, Holm-corrected p-values, so an improvement that sits inside the noise does not get reported as a win.
- **Maintain nine PyPI packages at 84–100% coverage**, CI-enforced via GitHub Actions and Codecov; **four** of the group's projects are built on the *exordium* feature-extraction toolkit.
- **Implemented the video processing and object tracking** under project EFOP-3.6.3-VEKOP-2017-00002, published as **two IJCNN (CORE A)** papers.

Associate Scientist, Institute for Computer Science and Control (SZTAKI) Jul 2014 – Aug 2017
Hungarian Academy of Sciences · Budapest, Hungary

- **Developed and published AHFS**, an adaptive hybrid feature selection method, in **Pattern Recognition (Q1)**: on the paper's benchmark it roughly **halves the model error** of the individual selectors it combines, on industrial data up to 1,100 variables wide. Deployed on wind-turbine monitoring and a machining situation-detection model that **runs in the machines' control systems**; since reused in clinical EEG and actigraphy screening, where it was the **best-performing** selector at 0.89–0.92 accuracy.
- **Ran a basic research project for OPEL Szentgotthárd** under iKOMP: feature selection and production trend forecasting on shop-floor data, which the plant used to bring engine failures down.
- **Built the institute's model-training and visualization framework**, extending its earlier tooling with data mining support; **SZTAKI adopted it** as the successor to their in-house system.

EDUCATION

PhD in Computer Science, summa cum laude Sep 2019 – Jun 2026
Eötvös Loránd University (ELTE)
Thesis: Affective Behavior and Social Interaction Analysis
Supervisor: Dr. habil. András Lőrincz

MSc in Computer Science, excellent Feb 2016 – Jul 2018
Eötvös Loránd University (ELTE)
Thesis: Speech de-identification with deep neural networks
Supervisor: Dr. habil. András Lőrincz

BSc in Computer Science, excellent Sep 2012 – Feb 2016
Eötvös Loránd University (ELTE)
Thesis: User interface design and implementation for data mining algorithms in MATLAB
Supervisor: Dr. Zsolt János Viharos

SKILLS

ML & Deep Learning:	PyTorch, Lightning, Transformers, computer vision, multimodal learning, multi-task learning, distributed training (DDP, multi-GPU), mixed-precision training
ML Systems & Optimization:	ONNX Runtime, model quantization & optimization, pruning, inference optimization
MLOps & Deployment:	MLflow, Optuna, Docker, Google Cloud Run, FastAPI, Gradio, Hugging Face Hub & Spaces, GitHub Actions, Codecov
Data & Pipelines:	NumPy, pandas, SQLAlchemy, HDF5, Hugging Face Datasets, video decoding, reproducible data pipelines, dataset unification & balancing
Engineering:	Python, Bash, Linux, Git, Apptainer, uv, Make, test-driven development, mypy, ruff, semantic versioning, Sphinx
Evaluation:	experimental design, statistical hypothesis testing, bootstrap confidence intervals, multiple-comparison correction, ablation studies

SELECTED PROJECTS

- BlinkLinMult** · Transformer-based eye blink detection and eye state recognition.
Production-oriented pipeline from training to real-time inference and deployment, separating PyTorch development from ONNX Runtime release with numerical parity checks on every export. Profiling and optimization reduced latency from **94 ms/frame to 23 ms/frame (42.8 fps)**. Versioned package with REST API, Docker image and live demo; **1,767 tests, 96% coverage**, evaluation on held-out recordings with controlled statistical comparisons.
PyTorch · ONNX Runtime · FastAPI · Docker · Gradio · Hugging Face Spaces
- PersonalityLinMult** · Multimodal Big Five personality perception, trained and served.
Multimodal model for Big Five personality perception from acoustic, visual and textual inputs. Optimized for CCC instead of MAE to avoid collapse to mean predictions; monitoring prediction spread exposed and resolved this issue. ONNX Runtime export achieved **6–8× faster** inference with **0.001–0.004** accuracy drift per trait. Packaged with MLflow tracking, Optuna tuning, Docker image and live demo; **84% test coverage**.
PyTorch · MLflow · Optuna · ONNX Runtime · FastAPI · Docker · Cloud Run
- Annie** · Local-first browser UI to explore, inspect and validate a video annotation dataset.
Browser-based UI for **human-in-the-loop dataset curation**, enabling expert review and correction of detections, tracks and segment boundaries to improve dataset quality. Runs locally with no cloud by design, keeping all data on-device to prioritize privacy. **346 tests, 96% coverage**, released to PyPI and Docker Hub.
Python · NiceGUI · torchcodec · SQLite · Docker

SELECTED PUBLICATIONS

- B. C. Dos Santos Melício, K. Karaköse, Á. Fodor, L. Xiang, V. Varga, L. Soorya, E. Dillon, P. Kun, A. Sárkány, M. Chetouani, K. Fenech. **Multimodal Framework for Automatic Behavior Analysis of Children with Autism During ADOS-2**. *Cognitive Computation*, vol. 17, 137, 2025. (Q1)
- Á. Fodor, K. Fenech, A. Lőrincz. **BlinkLinMult: Transformer-Based Eye Blink Detection**. *MDPI Journal of Imaging*, vol. 9-10, 196, 2023. (Q2)
- Z. J. Viharos, K. B. Kis, Á. Fodor, M. I. Büki. **Adaptive, hybrid feature selection (AHFS)**. *Pattern Recognition*, vol. 116, 107932, 2021. (Q1)

Full list of ten publications at adamfodor.com

ADDITIONAL INFORMATION

Awards:	1st place, Students' Scientific Conference (TDK), 2018 Special award, Morgan Stanley, 2018 Student TDK supervision award, 2023
Mentoring:	Supervised 2 BSc/MSc research projects and taught MSc courses over 8 semesters.
Languages:	Hungarian: native English: fluent
References:	Dr. habil. András Lőrincz · Founder of NIPG, ELTE · lorincz@inf.elte.hu Dr. Kristian Fenech · Head of NIPG, ELTE · fenech@inf.elte.hu